

Questions with Answer Keys

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Q1 - Sine Rule and Its Application

In ΔPQR , if $\sin P \sin Q = \frac{pq}{r^2}$, then R is equal to

(1) 45°

(2) 90°

(3) 60°

(4) 120°

Q2 - Sine Rule and Its Application

The value of $a^2 (\cos^2 B - \cos^2 C) + b^2 (\cos^2 C - \cos^2 A) + c^2 (\cos^2 A - \cos^2 B)$ is

(1) 0

(2) 2

(3) abc

(4) $a^2 b^2 c^2$

Q3 - Sine Rule and Its Application

If the angles of a triangle are in the ratio 4 : 1 : 1, then the ratio of the largest side to the perimeter is

(1) $\sqrt{3} : (2 + \sqrt{3})$

(2) 1 : 6

(3) $1 : (2 + \sqrt{3})$

(4) 2 : 3

Q4 - Sine Rule and Its Application

If in ΔABC , $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A-B)}{\sin(A+B)}$, then the triangle is

(1) right angled

(2) isosceles

(3) right angled or isosceles

(4) right angled isosceles

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Q5 - Cosine Rule and Projection Formulae

In $\triangle ABC$, if $\frac{\cos A}{b \cos C + c \cos B} + \frac{\cos B}{c \cos A + a \cos C} + \frac{\cos C}{a \cos B + b \cos A} = \frac{a^2 + b^2 + c^2}{\lambda abc}$ then the value of $\lambda + \lambda + \lambda + \dots$ upto 20 terms is

(1) 10

(2) 20

(3) 30

(4) 40

Q6 - Cosine Rule and Projection Formulae

In $\triangle ABC$, if $(a + b + c)(a - b + c) = 3ac$, then

(1) $\angle B = 60^\circ$

(2) $\angle B = 30^\circ$

(3) $\angle C = 60^\circ$

(4) $\angle A = \angle C = 90^\circ$

Q7 - Cosine Rule and Projection Formulae

In $\triangle ABC$, $\frac{2 \cos A}{a} + \frac{\cos B}{b} + \frac{2 \cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$, then the value of angle A is

(1) 45°

(2) 30°

(3) 90°

(4) 60°

Q8 - Cosine Rule and Projection Formulae

In $\triangle ABC$, if $A + C = 2B$, then $\frac{a+c}{\sqrt{a^2+c^2-ac}}$ is equal to

(1) $2 \cos\left(\frac{A+C}{2}\right)$

(2) $\sin\left(\frac{A+C}{2}\right)$

(3) $\sin \frac{A}{2}$

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(4) None of these

Q9 - Cosine Rule and Projection Formulae

If the line segment joining the points $A(a, b)$ and $B(c, d)$ subtends an angle θ at the origin, then $\cos \theta$ is

(1) $\frac{ab+cd}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

(2) $\frac{ac+bd}{\sqrt{(a^2+b^2)(c^2+d^2)}}$

(3) $\frac{ac+bd}{\sqrt{(a^2+c^2)(b^2+d^2)}}$

(4) None of these

Q10 - Napier Analogy and Half Angle Formulae

In $\triangle ABC$, the value of $1 - \tan \frac{B}{2} \tan \frac{C}{2}$ is

(1) $\frac{2a}{a+b+c}$

(2) $\frac{a}{a+b+c}$

(3) $\frac{2}{a+b+c}$

(4) $\frac{4a}{a+b+c}$

Q11 - Napier Analogy and Half Angle Formulae

In $\triangle ABC$, if $\sin \frac{B}{2} \cdot \sin \frac{C}{2} = \frac{b+c-a}{a}$, then the triangle is

(1) equilateral

(2) isosceles

(3) right angled

(4) not possible

Q12 - Napier Analogy and Half Angle Formulae

In $\triangle ABC$, $\sin \frac{A}{2} \sin \frac{B}{2} = \sin \frac{C}{2}$ and $2s$ is the perimeter of the triangle, then s is equal to

(1) $2c$

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- (2) c
- (3) $3c$
- (4) $4b$

Q13 - Napier Analogy and Half Angle Formulae

In a $\triangle ABC$, if $\cot \frac{A}{2} \cot \frac{B}{2} = c$, $\cot \frac{B}{2} \cot \frac{C}{2} = a$, $\cot \frac{C}{2} \cot \frac{A}{2} = b$. Then, the value of $\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c}$ is

- (1) 1
- (2) 2
- (3) 3
- (4) None of these

Q14 - Napier Analogy and Half Angle Formulae

In a $\triangle ABC$, $\tan \frac{A}{2} = \frac{5}{6}$ and $\tan \frac{C}{2} = \frac{2}{5}$, then

- (1) a, c, b are in AP
- (2) a, b, c are in AP
- (3) b, a, c are in AP
- (4) a, b, c are in GP

Q15 - Area of Triangle

In $\triangle ABC$, side $a = 4$ cm, $\angle B = 30^\circ$ and $\angle C = 45^\circ$, then the area of the triangle is

- (1) $\left(\frac{\sqrt{3}+1}{3}\right) \text{ cm}^2$
- (2) $4(\sqrt{3}-1) \text{ cm}^2$
- (3) $\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) \text{ cm}^2$
- (4) None of these

Q16 - Area of Triangle

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If in $\triangle ABC$, the altitudes p_1, p_2, p_3 are in HP which are drawn from the vertices A, B, C on opposite sides, then $\sin A, \sin B, \sin C$ are in

- (1) AP
- (2) GP
- (3) HP
- (4) None of these

Q17 - Area of Triangle

The area of $\triangle ABC$ is $20\sqrt{3} \text{ cm}^2$, $\angle C = 60^\circ$ and its perimeter is 20 cm, then the side c will be

- (1) 4 cm
- (2) 7 cm
- (3) 8 cm
- (4) None of these

Q18 - Area of Triangle

In $\triangle ABC$, $a^2 + b^2 = 1999c^2$, then $999 \left(\frac{\cot A + \cot B}{\cot C} \right)$ is equal to

- (1) 0
- (2) 1
- (3) 2
- (4) None of these

Q19 - Area of Triangle

If Δ is the area of triangle and $2s$ is the sum of three sides of a triangle such that $\Delta \leq \frac{s^2}{\lambda\sqrt{3}}$, then λ is equal to

- (1) 1
- (2) 2
- (3) 3
- (4) None of these

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Q20 - Properties of Triangles

In right angled $\triangle ABC$, if $\angle B = 90^\circ$, $a = 6$ cm and the radius of the circumcircle is 5 cm, then the area of $\triangle ABC$ is

- (1) 24 cm^2
- (2) 30 cm^2
- (3) 36 cm^2
- (4) None of these

Q21 - Properties of Triangles

If 3, 4, 5 be the sides of a $\triangle ABC$, then its inradius is

- (1) $\sqrt{\frac{8}{7}}$
- (2) $\sqrt{8}$
- (3) $\sqrt{7}$
- (4) None of these

Q22 - Properties of Triangles

In $\triangle ABC$, if $a = 5b = 12$ and $\angle C = 90^\circ$, then R is

- (1) $\frac{a}{2 \sin B}$
- (2) $\frac{b}{3 \sin B}$
- (3) $\frac{C}{\sin C}$
- (4) None of these

Q23 - Properties of Triangles

In $\triangle ABC$, $2R^2 \sin A \sin B \sin C$ is equal to

- (1) S^2
- (2) $ab + bc + ca$
- (3) Δ

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(4) None of these

Q24 - Properties of Triangles

In any $\triangle ABC$, the value of $a \cot A + b \cot B + c \cot C$ is

(1) $r + R$

(2) $r - R$

(3) $2(r + R)$

(4) $2(r - R)$

Q25 - Properties of Triangles

Let p, q, r be the sides of a $\triangle PQR$ such that $p : q : r = 4 : 5 : 6$, then the ratio of radius of circumcircle to that of incircle is

(1) $16 : 7$

(2) $7 : 16$

(3) $5 : 16$

(4) None of these

Q26 - Properties of Triangles

The sum of radii of inscribed and circumscribed circles for an n -sided regular polygon of side a , is

(1) $\frac{a}{2} \cot \frac{\pi}{2n}$

(2) $\frac{a}{2} \tan \frac{\pi}{2n}$

(3) $a \cot \frac{\pi}{2n}$

(4) None of these

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